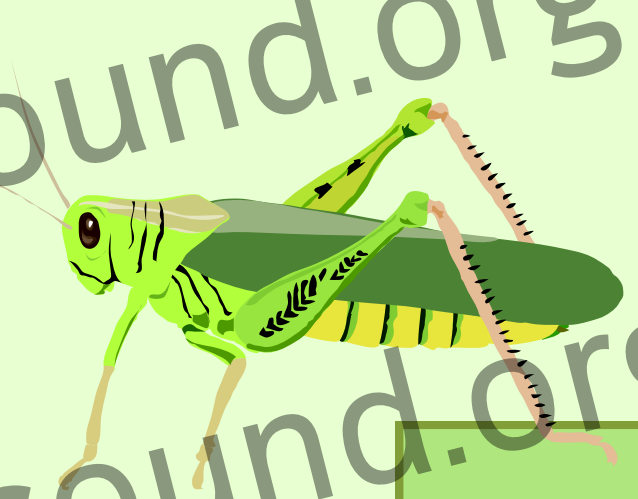
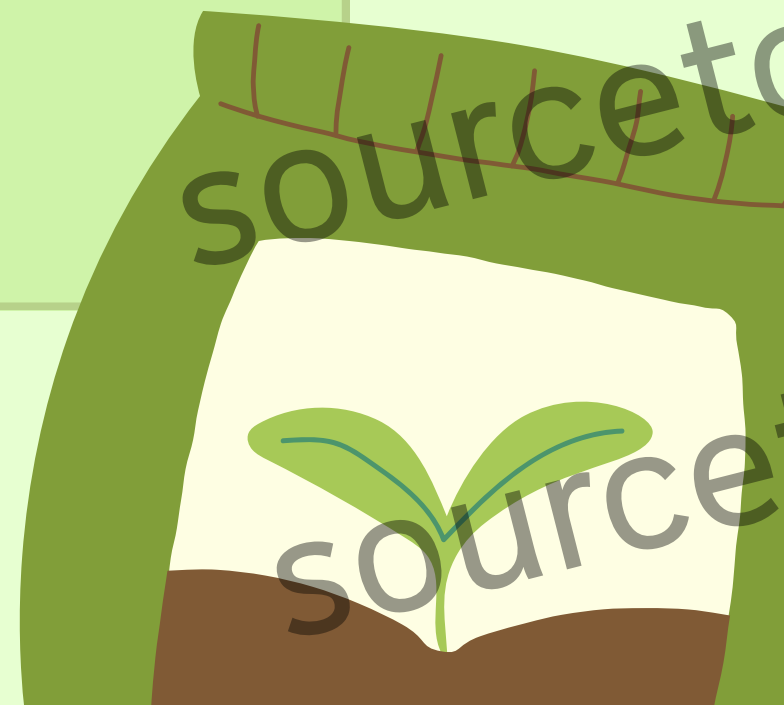




MACROINVERTEBRATE MEASUREMENTS

What can tiny stream animals tell us
about water health?



WHAT'S LIVING IN A STREAM?



WHAT IS A MACROINVERTEBRATE

1. **MACRO-**: large enough to see without a microscope

2. **INVERTEBRATE**: an animal without a backbone

MACROINVERTEBRATE

animals without a backbone large enough to be seen with the naked eye



A



B



C



F



E



D



MEET SOME BIOINDICATORS!

Group 1 (Sensitive)	Group 2 (Moderately Tolerant)	Group 3 (Tolerant)
Mayfly nymph	Dragonfly nymph	Aquatic worm
Stonefly nymph	Aquatic beetle	Midge larva
Caddisfly larva	Crayfish	Leech

GROUP 1 (SENSITIVE)

Requirements: Need clean, cool, oxygen-rich water and generally cannot tolerate much pollution

Mayfly nymph	Water Penny Beetle	Caddisfly larva
		
immature stage of stonefly insect; usually lives under rocks in fast-moving streams	Small, oval-shaped aquatic insects that resemble tiny pennies; cling to undersides of rocks and feed on algae	aquatic insect larva; many build protective cases from sticks or rocks

GROUP 2 (MODERATELY TOLERANT)

Requirements: Can handle some environmental change or pollution, stil needs suitable habitats

Dragonfly nymph	Aquatic Beetle	Crayfish
		
aquatic juvenile of a dragonfly and an active predator; hides among plants, mud, and debris	Spends part or all of its life in water; lives among aquatic plants and rocks	freshwater crustacean related to lobsters; hides under logs and stream-bank crevices

GROUP 3 (POLLUTION-TOLERANT)

Requirements: Can survive lower oxygen, warmer water, and more disturbed conditions than sensitive organisms

Aquatic worm	Midge larva	leech
		
Small segmented worm that lives in bottom sediment; can survive in muddy areas with low oxygen	worm-like organisms that thrive in low-oxygen, nutrient-rich water; often called "bloodworms" for red appearance	soft-bodied, segmented worm; lives in slow water, muddy bottoms, areas with abundant organic material

YOUR MISSION

1. **FIND:** the organisms hiding in your stream
2. **IDENTIFY:** use your guide to figure out what you found
3. **CLASSIFY:** sort each macroinvertebrate by pollution tolerance
4. **INVESTIGATE:** Use your evidence to determine stream health

DISCUSSION

Let's Compare Our Streams!

	Stream 1	Stream 2	Stream 3
Sensitive Organisms			
Moderately tolerant			
Tolerant			
Water Quality			

WRAP-UP: WHAT CAN BUGS TELL US?

What we learned:

- **Macroinvertebrates** are small animals without backbones that we can see without a microscope
- **Bioindicators** are living organisms that help scientists understand environmental health
- Scientists can study the **types and diversity** of organisms to determine **water quality** in a stream



EVEN THE SMALLEST CREATURES CAN TELL USE BIG THINGS!